

Annotation Versioning and Cross-Collection Generalization in Varroa Mite Counting

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Abstract

Reliable Varroa mite counting requires evaluation protocols that distinguish annotation effects, shared image content, and transfer across acquisition sources. This study examines these factors through a paired annotation audit, spatial-overlap screening, inference controls, and a frozen-checkpoint evaluation of a YOLO11n detector on a second public collection. The first dataset contains 64 photographs and two annotation representations: 807 CSV boxes and 849 XML boxes. All 42 additional XML boxes occur in the 14-image validation partition. Holding the model, predictions, and operating threshold fixed, changing the reference from CSV to XML changes AP50 from 0.8981 to 0.9130, precision from 0.7345 to 0.8276, and mean absolute count error from 4.64 to 3.36 per photograph. Geometric matching and visual inspection identify shared scene regions in three training-validation image pairs. An additional collection contains 180 crops with 8,572 annotations and 25 filename-derived parent groups; 22 groups span the uploader's partitions. On this collection, the unchanged detector produces no detections at the prespecified confidence threshold of 0.25. A known positive source tile reproduces its cached confidence exactly, providing a runtime control. The contribution is a reproducible empirical account of how annotation representation, sampling units, and collection transfer constrain the interpretation of a detection benchmark. The findings apply to the evaluated checkpoint and releases; independent field performance remains unestablished.

Keywords: Varroa destructor; object detection; dataset audit; annotation versioning; domain shift; reproducibility.

1. Introduction

Image-based counting can reduce the manual effort required to inspect Varroa mite fall on sticky boards. Reliable evaluation requires more than a detector score: the counted region, annotation representation, acquisition unit, and relation between training and evaluation images determine what that score means. A photograph may depict only part of a board, and two distinct files can contain overlapping scene regions. Likewise, a dataset distributed in several annotation formats may not define a single interchangeable reference.

Divasón et al. released high-resolution sticky-board photographs, annotations, and detection software [1,2]. Their work provides a valuable basis for reproducible experiments. Subsequent VarroDetector software uses YOLO11 and an explicit counting region [3]. Slicing-based detection is already established both in Varroa applications and in general small-object detection [1,4]. We therefore investigate evaluation reliability rather than presenting tiling or YOLO11 as a new algorithm.

The research questions are: (i) how much do two distributed annotation representations change the evaluation of identical predictions; (ii) does file-level separation ensure scene-level separation in the supplied split; (iii) how do input scale and cross-tile duplicate suppression affect the same checkpoint; and (iv) does its validation performance transfer to a separately hosted public collection without adaptation? These questions concern the relationship between measured performance and scientific claims, consistent with broader work on leakage and reproducibility [5].

The study contributes three linked empirical results. First, paired evaluation of frozen predictions quantifies the effect of annotation representation separately from changes in the detector. Second, image-level and geometric checks expose sampling dependencies that exact-duplicate screening misses. Third, a protocol fixed before additional-collection inference measures transfer failure and checks it against a positive runtime control.

Versioned references, per-image predictions, and portable evaluation code make these measurements inspectable and reproducible. The contribution is specific to the audited releases and checkpoint; published detectors evaluated under different protocols are not ranked against it.

1.1 Selected related work

Visual Varroa research includes detecting mites on bee bodies and counting fallen mites on collection boards. These tasks differ in background, scale, occlusion, and the meaning of a counted instance. Bilík et al. compared object detection and anomaly detection on images of healthy and infected bees [13]. Divasón et al. addressed sticky-board analysis using Faster R-CNN, image slicing, and refinement [1]. The later VarroDetector study combined YOLO11 with an explicit counting region and a larger collection, including a separate evaluation on newly acquired sticky sheets [3]. Table 1 places the present study within these selected contributions; it is not an exhaustive literature review.

Study	Data and observation unit	Approach and evaluation	Reported outcome
Bilík et al., 2021 [13]	600 images of healthy and infected bees	YOLO, SSD, and Deep SVDD; infected-bee and mite recognition	F1 up to 0.874 for infected bees and 0.727 for mites
Divasón et al., 2024 [1]	64 sticky-board photographs; 50 training and 14 validation	Faster R-CNN, slicing, refinement, and optional deblurring	Best reported AP50 0.907; mAR100 0.967
Yániz et al., 2025 [3]	357 photographs from 90 sheets; separate evaluation on 114 newly acquired sheets	YOLO11n and counting-region extraction; validation and counting evaluation	Validation precision 0.925, recall 0.921, and reported mAP 0.956
This study	The 64-photograph release, two references, and 180 additional crops	One YOLO11n checkpoint; annotation audit, overlap checks, inference controls, and frozen transfer	Reference-dependent source AP50 0.8981 / 0.9130; additional-collection AP50 0.000900

Table 1. Selected research context. Results retain each publication's task and evaluation conditions. The 2024 paper explicitly defines its reported mAP at IoU 0.5; the 2025 mAP is left as reported rather than reassigned an unverified IoU definition. Different data, references, sampling units, and protocols prevent a head-to-head performance ranking. The present source validation was used for checkpoint selection.

The open datasets and implementations in this literature enable scrutiny of an experimental pipeline beyond a single aggregate score. Our question is whether the interpretation of that score survives changes in released annotation representation and acquisition collection. We do not propose a new detection architecture. The added evidence consists of paired reference measurements, spatial checks of the supplied split, and an explicitly frozen transfer result.

2. Materials and provenance

2.1 High-resolution sticky-board collection

The Zenodo release by Divasón and colleagues [2] contains 64 RGB photographs, each 6048 pixels wide and 8064 pixels high, distributed under CC BY 4.0. The supplied CSV assigns 50 photographs to training and 14 to validation. It represents 580 training and 227 validation boxes. The XML files represent 580 training and 269 validation boxes. We preserve both references and the supplied image partition.

The source archive was verified by SHA-256 against the acquisition receipt. Every image and XML file was also checked against the retained manifest. XML coordinates were converted by subtracting one from all four endpoints, matching the inspected original evaluation notebook at commit 6dabb68f4580bd9224224bd8f7573a2be37a2bf6 [9]. This includes preserving boundary coordinates of -1 where they occur; we did not silently clip or relabel the source reference. We then verified every reconstructed XML box against the actual archived XML.

The photograph is the counting unit in this study. Physical-board, colony, and capture-session identifiers were not independently verified. The source publication discusses green strings defining a counting region [1]; our whole-photograph counts are not sums over distinct physical boards and are not colony infestation estimates.

2.2 Additional collection

We acquired version 4 of the public Roboflow project titled varroa by the account Tomislav uri [6]. Its export README and COCO metadata specify CC BY 4.0. The archive contains 180 images of 1024 by 1024 pixels and 8,572 boxes. The uploader's partitions contain 126, 36, and 18 images. No image is empty; the version page specifies removal of images without annotations. The maximum count is 215, and 24 images contain more than 100 annotations.

Although this export version lists no augmentation or resizing, its filenames identify crops. We parse the numerical prefix preceding the crop index, obtaining 25 putative parent-image groups containing 3-12 crops each. Twenty-two groups occur in multiple uploader partitions. These are filename-derived groups, not validated identities of 25 physical boards. Original acquisition conditions and upstream parent images were not supplied. We therefore call this an additional-collection evaluation, not a verified independent-apiary trial.

All 180 crops are used only for evaluating the already selected checkpoint. The uploader's partitions are retained as provenance metadata but are not used to train a model in this study. No external label is used for checkpoint selection, confidence tuning, or detector adaptation. Automated screening cannot prove the absence of cross-source contamination.

Collection and role	Image files and resolution	Reference boxes	Sampling information
Source training [2]	50 photographs, 6048 x 8064	CSV: 580; XML: 580	Supplied training partition; CSV used for fitting
Source validation [2]	14 photographs, 6048 x 8064	CSV: 227; XML: 269	Checkpoint-selection partition; not an independent test
Additional evaluation [6]	180 crops, 1024 x 1024	COCO: 8,572	25 filename-derived parent groups; physical-board identities unknown

Table 2. Data used in the experiments. CSV and XML describe the same source photographs, so their box counts are alternative references, not additive observations. Additional-collection crops are not treated as 180 independent boards. Both releases specify CC BY 4.0.

3. Methods

3.1 Detector training and reconstruction

We trained a single YOLO11n detector [7] on EVREN [8], using the first collection's CSV labels and supplied split. EVREN describes itself as an end-to-end AI platform developed within Türkiye's Presidency of Defence Industries (SSB) [12]. We used it for training and model sharing; dataset preparation, provenance checks, and ONNX evaluation ran locally. Images were divided into 1024-pixel square tiles with nominal stride 896; the last tile was anchored to the image edge. A source photograph never contributed tiles to both training and validation. This construction prevents the same source file from crossing partitions but does not by itself prevent shared scenes across distinct files.

Training used 569 positive and 569 deterministically sampled negative tiles. All 882 validation tiles were retained, including 672 without CSV annotations. The training and validation tiles contained 838 and 325 box appearances, respectively; overlapping tile appearances are not unique mites. All 807 source CSV boxes were fully visible in at least one tile. Tiles were encoded as JPEG at quality 95 with chroma subsampling disabled, and their hashes were frozen.

The EVREN configuration used image size 1024, batch size 16, seed 42, a maximum of 40 epochs, patience 10, and one GPU. Light color, translation, scale, rotation, and horizontal-flip augmentation was enabled; mosaic,

mixup, copy-paste, and erasing were disabled. The platform recorded completion after 33 epochs. The exact best-checkpoint epoch and GPU model were not independently established. The exported ONNX metadata reports Ultralytics 8.3.0.

Inference used ONNX Runtime 1.30.0 on CPU, with four intra-operation threads. Inputs were RGB float32 values divided by 255. We retained candidate scores at least 0.001, applied per-tile non-maximum suppression (NMS) at IoU 0.7 with a limit of 300, translated boxes to source-image coordinates, and applied global NMS at IoU 0.5 with a limit of 1000. The operating confidence threshold was fixed at 0.25 before the original inference. The model is publicly accessible on EVREN [8].

3.2 Annotation and overlap audit

We compare exact coordinate sets and use one-to-one descending-IoU matching at IoU at least 0.5 to characterize annotation geometry. We count ambiguous candidate matches rather than presuming that every coordinate difference denotes a distinct object. Neither reference is treated as newly expert-adjudicated ground truth.

For spatial screening, we examined all 2,016 unordered pairs of source photographs. ORB features were extracted at a longest image edge of 1024 pixels; ratio-filtered descriptor matches were fit with a RANSAC homography. Pairs with at least 20 inliers and an inlier ratio of at least 0.35 were retained as candidates. This is an exploratory retrieval screen, not a validated duplicate classifier. Its candidates can arise from repetitive texture or reused photographic equipment.

The three cross-partition candidates were checked using SIFT at a longest edge of 2048 pixels, a descriptor ratio threshold of 0.7, and a three-pixel RANSAC tolerance. Dark, green, and outer-border keypoints were masked. An initial capped-feature diagnostic was poorly conditioned on narrow overlap; the final evidence uses uncapped SIFT features and retains this analytical revision in the record. Registered images were visually inspected by the AI assistant. This is not independent human annotation review or confirmation by the data creators. The implementation follows standard feature-matching and homography operations [10].

The second collection was screened against the first through 11,520 ORB comparisons. Eight candidates underwent a second SIFT check with unique destination correspondences and visual inspection. We do not interpret failure to find a supported match as proof of distinct provenance.

3.3 Evaluation and inference controls

We report COCO-style bounding-box AP at IoU 0.50 and averaged over 0.50:0.05:0.95 [11], alongside fixed-threshold TP, FP, FN, precision, recall, mean absolute count error (MAE), root mean squared count error (RMSE), and signed bias. Each photograph or exported crop contributes one count observation. With predicted count p and reference count g , MAE is the mean of $\text{abs}(p-g)$, RMSE is the square root of the mean squared difference, and bias is the mean signed difference $p-g$.

Source AP uses a maximum of 100 detections per image. No source image exceeds that ceiling at confidence 0.25. Since the additional collection includes dense crops, we report both the standard AP ceiling of 100 and an expanded ceiling of 1000, and use the larger ceiling for fixed-threshold matching. Precision is undefined when there are no predictions. We do not treat correlated crops as independent biological replicates, report population confidence intervals, or perform significance tests based on unverified sampling units.

Three paired source-image conditions use exactly the same weights: (a) the complete photograph resized with aspect ratio preserved to a 1024-square input with centered value-114 padding; (b) tiled predictions without cross-tile suppression; and (c) tiled predictions with global NMS. Each is evaluated against CSV and XML references. These are post-hoc controls on the existing validation set, not independent tests or a comparison against a separately trained full-image detector.

The additional-collection protocol and model hash were saved before model inference, after admission checks and before its prediction scores were examined. Its 1024-square images were re-encoded with the same JPEG settings as training tiles, without geometric resizing. We retained the prescribed tile and global NMS operations.

After observing transfer failure, a source tile with a previously cached positive response was reconstructed byte-for-byte and run through the same runtime as a positive control. The operating threshold was not changed.

3.4 Public trained model and experiment identity

The EVREN record titled Varroa Tile1024 YOLO11n Baseline [8] was publicly marked open and experimental on 23 September 2026. Its version panel lists v1.0.0 and provides an ONNX download. The local experiments use that exported artifact: 10,422,113 bytes, one class named varroa_mite, and SHA-256 5787ed2e84612aba615a2238218a50d98f64317c3cbf0d1d7550355dd2b21b03. The public model card specifies AGPL-3.0. The companion package identifies the checkpoint but does not redistribute its weights.

The platform's version summary reports mAP50 of 88.2% and mAP50-95 of 42.9%. These platform-reported values are separate from our local, photograph-level reconstruction and its two references. We have not established identical evaluation semantics for the platform summary, so those values are provenance information rather than substitutes for Table 3. All six source conditions use this one checkpoint; they are not six independently trained models. Public model and source-data links are provided in reference [8].

4. Results

4.1 Annotation representation changes measured performance

Ten of the 64 source photographs have at least one coordinate or count difference between references. Exact coordinate matches number 576 in training and 188 in validation. At IoU at least 0.5, all 580 training and 227 validation CSV boxes have an unambiguous XML counterpart. The remaining 42 XML boxes are all in validation. Thus the discrepancy combines coordinate revisions with additional represented instances; it is not a uniform coordinate-origin offset.

For the identical 290 detections above confidence 0.25, CSV evaluation yields 213 TP, 77 FP, and 14 FN; XML evaluation yields 240 TP, 50 FP, and 29 FN. Precision changes from 0.7345 to 0.8276, while recall changes from 0.9383 to 0.8922. AP50 increases from 0.8981 to 0.9130, whereas AP50-95 changes only from 0.4582 to 0.4574. MAE changes from 4.64 to 3.36 and signed bias from +4.50 to +1.50. These are changes in evaluation reference, not improvements in the detector.

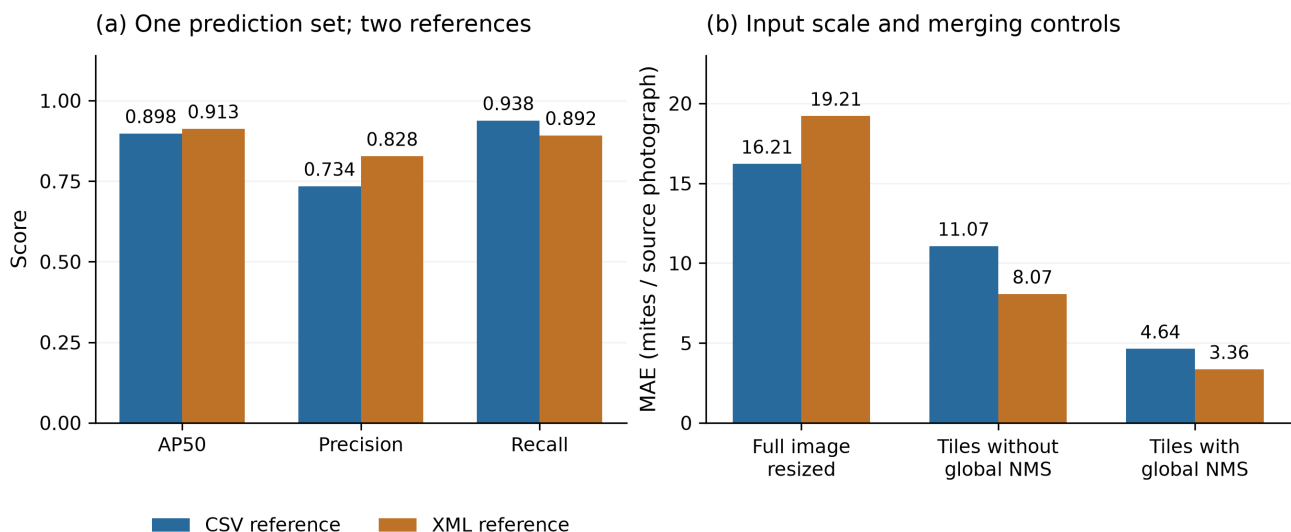


Figure 1. Paired annotation and inference-control results on the same 14 source validation photographs. (a) AP50 uses the retained score ranking; precision and recall use confidence 0.25. Changing the reference modifies the evaluation of an unchanged prediction set. (b) Count MAE for the same checkpoint under full-image resizing and the two tiled conditions described in Section 4.3. Bars are descriptive values, not estimates from independent training repeats. CSV and XML are alternative references for the same photographs.

4.2 Scene overlap crosses the supplied partition

There are no exact file or pixel duplicates among the 64 photographs. The geometric screen retrieves 24 candidate pairs, three of which cross training and validation. Secondary SIFT matching supports shared scene regions between IMG_5584 and IMG_5585 (57 inliers), IMG_5770 and IMG_5771 (106), and IMG_5907 and IMG_5908 (42). Median inlier reprojection errors are approximately 0.72, 0.91, and 1.57 pixels at a longest edge of 2048, respectively. Registered debris patterns provide visual evidence beyond identical filenames or a similar white background. Green counting strings can occur at different positions relative to the matched debris.

This establishes a limitation of interpreting the supplied file split as scene-disjoint. It does not quantify score inflation or show that every pair in the candidate list shares a physical board. Removing these three validation images afterwards would not produce an independent test: all 14 images have already participated in checkpoint selection and subsequent analysis.

IMG_5770.jpg (val) | IMG_5771.jpg (train) | inliers:29/42

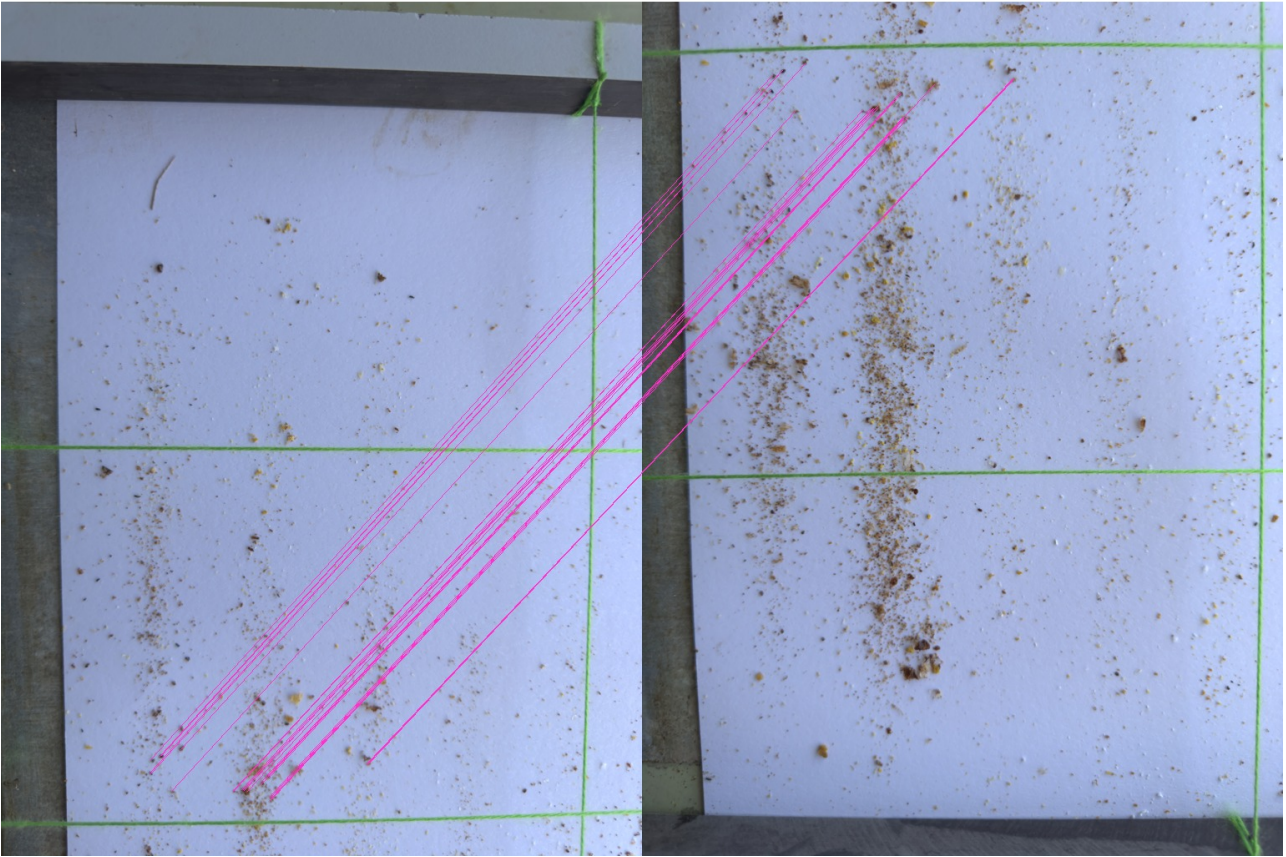


Figure 2. Shared debris patterns between IMG_5770 (validation, left) and IMG_5771 (training, right). Lines show ORB screen matches; the separate SIFT verification reports 106 inliers. Photographs from Divasón et al. [2], CC BY 4.0; correspondence overlay added for this audit. This figure documents overlap, not its effect on model performance.

4.3 Input scale and merging affect counting

Inference condition	Reference	Predictions at 0.25	AP50	AP50-95	MAE	Bias
Full image resized to 1024	CSV	0	0.0000	0.0000	16.21	-16.21
Full image resized to 1024	XML	0	0.0000	0.0000	19.21	-19.21
Tiles, without global NMS	CSV	382	0.7396	0.3806	11.07	+11.07
Tiles, without global NMS	XML	382	0.7369	0.3729	8.07	+8.07
Tiles, with global NMS	CSV	290	0.8981	0.4582	4.64	+4.50
Tiles, with global NMS	XML	290	0.9130	0.4574	3.36	+1.50

Table 3. Paired results on the same 14 validation photographs, using one checkpoint and a fixed operating confidence of 0.25 for counts. AP uses predictions retained at the lower score floor of 0.001. Counts refer to whole photographs rather than distinct physical boards. Figure 1b visualizes the count errors.

Global NMS removes 92 above-threshold detections while retaining the same TP count under each reference. The resizing control produces no above-threshold detections. Median source CSV box dimensions are 60 by 61 pixels, becoming approximately 7.62 by 7.75 pixels after full-image resizing. This control documents an input-scale mismatch for the tile-trained model; it cannot establish that full-image training is intrinsically ineffective.

4.4 Validation performance does not transfer to the additional collection

The unchanged detector produces zero predictions at confidence 0.25 across all 180 external crops. Consequently, TP=0, FP=0, FN=8,572, recall=0, and precision is undefined. MAE is 47.62 mites per crop, RMSE is 67.19, and bias is -47.62. All count errors represent undercounting to zero. AP50 is 0.000900 and AP50-95 is 0.000165 under either AP detection ceiling. Thus the near-zero ranking performance is not merely an artifact of the chosen operating threshold.

There are 921 retained predictions at the lower score floor of 0.001; their maximum score is 0.15968. The positive source control reproduces its cached maximum confidence exactly at 0.797905. The runtime therefore still produces the previously observed positive response on source data. This control does not isolate which changes in object scale, substrate, lighting, annotation style, or acquisition account for transfer failure.

The cross-source screen's eight candidates are not supported by secondary geometry or visual comparison: they yield only 0-6 SIFT inliers with unique correspondences, and exhibit different substrates and scene patterns. No crop was excluded before inference. Upstream acquisition and physical-board identities nevertheless remain unverified, so this result is an additional-collection stress test rather than an independent-apiary estimate.

Evaluation reference	Units	Precision	Recall	AP50	MAE	RMSE
Source CSV	14 photographs	0.7345	0.9383	0.8981	4.64	6.39
Source XML	Same 14 photographs	0.8276	0.8922	0.9130	3.36	5.24
Additional COCO	180 crops	Undefined	0.0000	0.000900	47.62	67.19

Table 4. Frozen-checkpoint results with global NMS. Precision and recall use confidence 0.25 and matching IoU 0.5. MAE and RMSE are mites per listed unit; photograph-level and crop-level errors do not measure the same counted region. Additional-collection precision is undefined because there are no operating-threshold predictions. AP50 is identical under ceilings of 100 and 1000 for that collection.

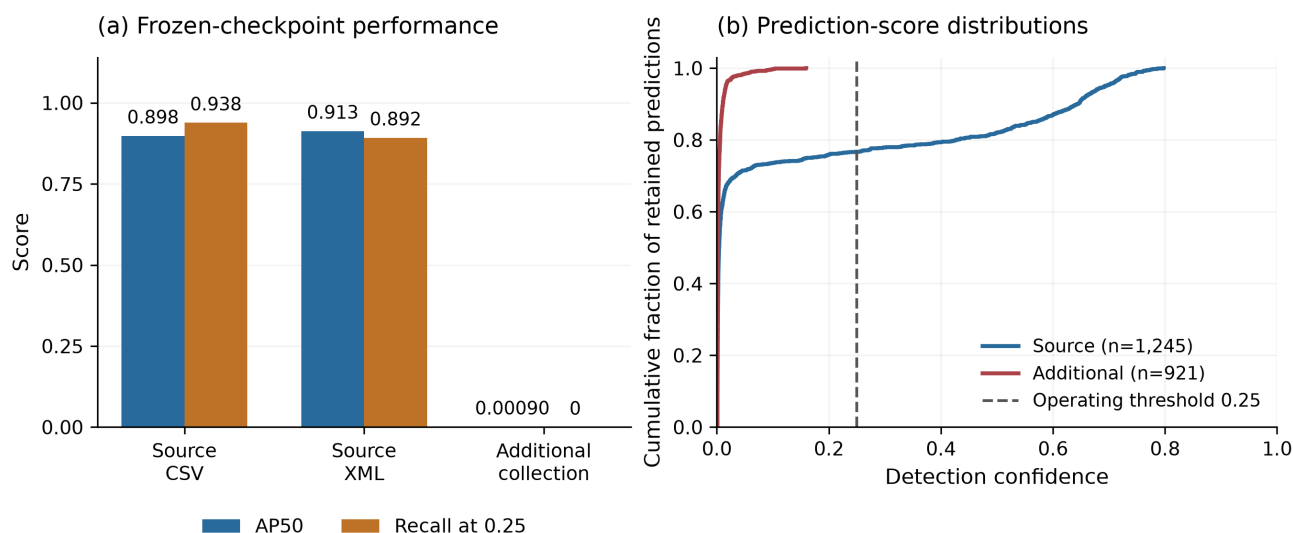


Figure 3. Descriptive source and additional-collection performance. (a) AP50 and fixed-threshold recall for the two source references and the additional collection. The source bars use the checkpoint-selection set; the additional collection is a transfer stress test. (b) Empirical cumulative distributions of retained detection scores, with the operating threshold marked. The distributions contain 1,245 source and 921 additional-collection predictions with scores at least 0.001 after global NMS. They describe retained predictions, not independent images or biological samples, and are not probability-calibration curves. All retained additional-collection scores are below 0.25.

5. Discussion

Annotation versioning changes both the numerator and denominator of common metrics. Here, XML evaluation converts some CSV false positives into matches while adding previously unrepresented targets, raising precision and lowering recall simultaneously. A headline AP50 improvement without a frozen prediction record would conceal that the detector did not change. Reporting the reference-file hash and per-image annotation counts is therefore essential to interpreting a comparison.

The overlap analysis shows why file-level deduplication is insufficient for high-resolution, partly overlapping photographs. Neither unique filenames nor unique pixel hashes establishes independent scenes. Our training preparation preserved source-image membership, yet inherited the supplied partition's scene overlap. A future scene-disjoint experiment requires confirmed acquisition groups or a clearly documented conservative grouping procedure and a new model fit. The present evidence must not be used to estimate the magnitude of contamination effects.

The additional collection also illustrates how dataset-page image counts can overstate sampling diversity. Its 180 images are crops assigned to only 25 filename groups, and most groups cross the uploader's partitions. We avoid that partition in model fitting, but cannot convert the filename count into a verified number of biological samples. Future dataset cards should distinguish files, crops, source photographs, physical boards, and colonies.

The substantial transfer failure limits the practical interpretation of the detector's source validation score. Lowering the confidence threshold after seeing the external result would change the question and require a separate development set. Instead, the frozen evaluation is retained. A prospective adaptation study should define target-domain development groups and untouched evaluation groups before fitting or choosing an adapted model, include empty boards, and evaluate both counting and detection. Increasing model capacity alone would not resolve uncertain labels or sampling units.

The comparison with previous research concerns evaluation design as much as detector choice. Our XML AP50 of 0.9130 must not be interpreted as an improvement over the 2024 paper's reported 0.907: annotation representation, inference pipeline, and selection conditions have not been held constant between studies [1]. Likewise, the larger training collection and separately acquired sheets in VarroDetector address evidence that the present experiment does not provide [3]. A useful extension would compare these open implementations

under a common, prospectively fixed reference and verified acquisition groups. Such a matched comparison has not been performed here.

The resulting experimental record supports an evaluation workflow in which annotation versions, shared scene content, and transfer performance are examined separately. Its practical value is the ability to trace a reported score to a specific reference and prediction set, and to distinguish an evaluation change from a model change. Testing this workflow on additional independently documented collections would establish how broadly the observed issues apply.

6. Limitations

The study uses one checkpoint from one training seed and does not compare detector families under matched training budgets. Source analyses are post-hoc and use the checkpoint-selection partition. Annotation differences have not been adjudicated by an independent Varroa expert. Geometric screening can miss overlap or retrieve false matches, and the scene review was performed with AI assistance rather than by an independent human panel. Actual physical-board and apiary identities remain unknown.

The additional collection has no empty crops and incompletely documented upstream provenance. Its labels are accepted as the released reference, not independently validated. Findings are conditional on the exact checkpoint, preprocessing, thresholds, and downloaded versions. CPU timing is descriptive and not a controlled hardware benchmark. We neither estimate colony infestation nor recommend biological treatment decisions from these counts.

7. Conclusion

An open Varroa counting pipeline can show high validation AP while its reported counting error depends on annotation representation and its performance fails to transfer to another collection. In this case, 42 additional validation XML boxes materially change the evaluation of fixed predictions; three image pairs show cross-partition shared scene regions; and a frozen EVREN-trained detector produces no operating-threshold detections on 180 additional crops. Versioned references, explicit sampling units, and separately reported transfer tests are necessary components of an interpretable benchmark. The next research step is a prospectively defined, group-aware adaptation study with independently reviewed labels and empty-board evaluation.

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Data, software, and AI assistance

Source data are available from Zenodo [2] and Roboflow [6] under CC BY 4.0; the EVREN model is public [8]. The companion package includes references, predictions, protocols, audit evidence, and three portable scripts for metric calculation, model inference, and quantitative figure generation. Figures 1 and 3 are generated from the recorded metric and prediction files, with source-file hashes retained. All eight metric configurations were reproduced. Rerunning all 14 source photographs and 180 additional crops produced numerically identical predictions on the recorded runtime. The package does not reconstruct training or every exploratory audit step. The preprint and companion package share the reserved Zenodo record identified above; the DOI becomes active upon publication.

OpenAI Codex was used for literature retrieval, source inspection, code generation and execution, experimental analysis, image inspection, and manuscript drafting. Model scores and counts reported here were computed from the described files and model inference; they were not generated as hypothetical results. AI assistance is not an independent scientific reviewer or author. Responsibility for the interpretation, accuracy, and submitted version rests with the named human author.

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